

Course Description

Prof. Dr. Gerardo Flores

Robotics and Automation (CSCE 3345 - SENG 3340)

TAMIU



Instructor Information

Gerardo Flores, Ph.D.
Associate Professor of Systems Engineering
Director, RAPTOR Lab
Texas A&M International University

Email: gerardo.flores@tamiu.edu

Office: LBV 303

To schedule an appointment, please contact me through Blackboard or email.

Robotics and Automation (CSCE 3345 - SENG 3340)

Study of the use, design, and deployment of industrial automation and robotics technologies in high-precision, multi-product manufacturing environments. Robot manipulators, kinematics and dynamics, robot automation and control, integrated robotic systems for manufacturing, automation in manufacturing, programmable logic controllers, applications to industrial systems. Interchangeable with CSCE 3345.

Prerequisite: ENGR 2305, ENGR 2105, and MATH 3310

Course Information:

•Course Overview

- Comprehensive introduction to robotics and automation principles.
- Focus on designing, analyzing, controlling, and implementing robotic systems.

•Key Topics

- **Robotic Manipulators:** Forward and inverse kinematics, Jacobians, dynamics, and motion planning.
- **Automation and Control:** Applying control theories for precise robotic operations.
- **Mobile Robotics:** Fundamentals of mobile and autonomous systems.

•Practical Integration

- Combines theory with hands-on experience using **Robotics Toolbox**, **Python**, and **Webots**.
- Simulation and control of robotic systems in virtual environments.

•Outcomes

- Develop problem-solving skills for academic and industrial challenges.
- Gain expertise in the simulation and implementation of robotic systems.

Prerequisites:

- Basic knowledge of **linear algebra**, **differential equations**, and ***control systems***.
- Familiarity with programming languages such as Python or C++ is recommended.

Target Audience

- Ideal for undergraduate and graduate students in engineering, computer science, and related fields.
- Focused on those interested in robotics, automation, and intelligent systems.

AI Tools

- Encouraged as supplementary resources for learning and problem-solving.
- **Not permitted during exams**; students will be evaluated on their own knowledge and skills.

Program Learning Outcomes

- Mathematical Modeling and Analysis**

- Proficiency in deriving and analyzing kinematic and dynamic models for manipulators and mobile robots.

- Control System Design**

- Design and implement simple controllers to ensure stability, accuracy, and efficiency.

- Simulation and Implementation**

- Utilize tools like Robotics Toolbox, Python libraries, and ROS to simulate, control, and analyze robotic systems.

- Problem-Solving and Critical Thinking**

- Solve complex engineering problems in robotics with innovative, systematic approaches.

- Programming and Software Development**

- Develop software for robotics applications, including sensor integration and real-time control in Python or C++.

- Collaborative Project Management**

- Work in multidisciplinary teams, demonstrating project management and communication skills.

- Innovation and Lifelong Learning**

- Stay updated on robotics and automation, fostering creativity and continuous learning.

Student Learning Outcomes

1. Develop a deep understanding of robotics' mathematical models and control algorithms.
2. Gain proficiency in using **Python** and simulating and controlling robotic systems.
3. Design and analyze robotic systems for various applications, including industrial automation and autonomous navigation.
4. Understand and address ethical issues and societal impacts of deploying robotics and automation technologies.

Important Dates

Visit the Academic Calendar (tamiu.edu) (<https://www.tamiu.edu/academiccalendar/>) page to view the term's important dates.

Students must also check **Blackboard** and pay attention to **class announcements** for upcoming exams, assignments, and projects (most already listed in the syllabus).

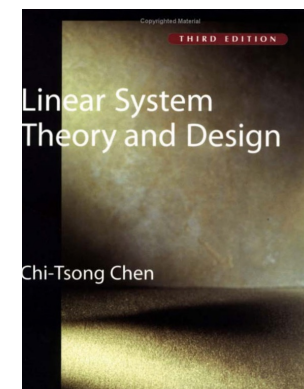
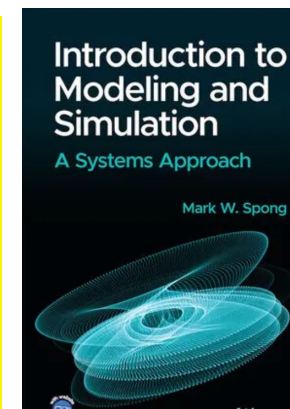
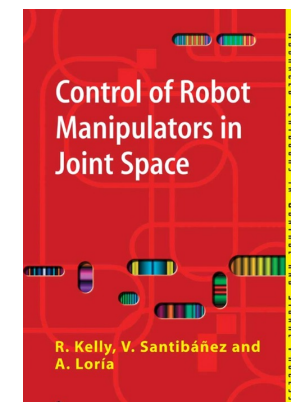
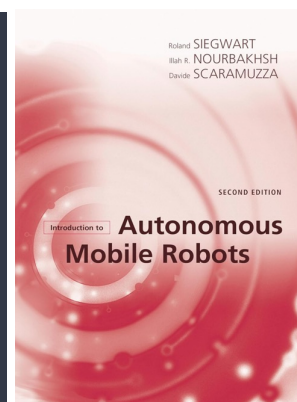
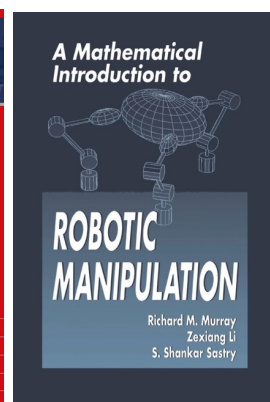
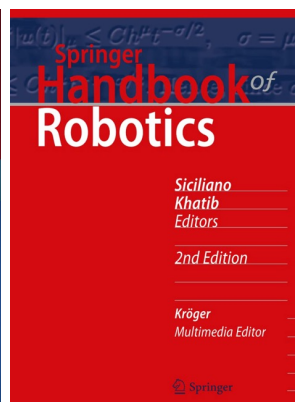
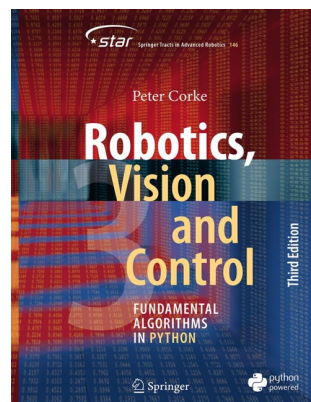
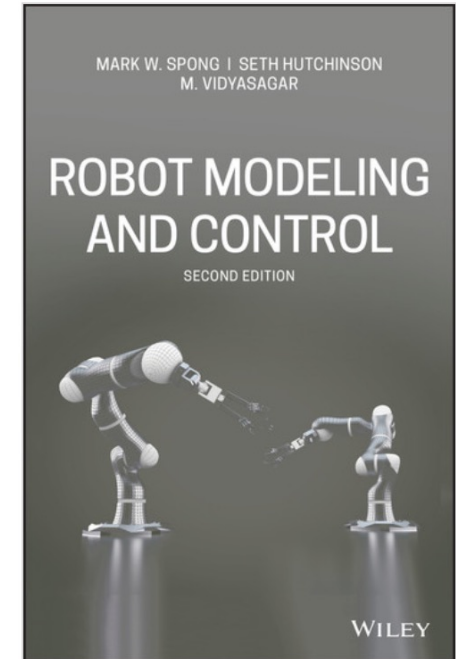
Course Contents

1. Introduction to Robotics and Fundamental Concepts
2. Control and Automation Fundamentals for Robotics
3. Rigid Motions and Homogeneous Transformations
4. Forward Kinematics
5. Inverse Kinematics
6. Velocity Kinematics and Jacobians
7. Robot Dynamics and Mathematical Modeling
8. Robot Control
9. Laboratory Experimentation and Implementation
10. Vision-Based Robot Control*

Note: A detailed tentative schedule, including dates and assignments, will be published in the course syllabus on Blackboard this week.

Textbooks

Group	Title	Author	ISBN
Optional	Robot Modeling and Control	Mark W. Spong, Seth Hutchinson, M. Vidyasagar	978-1-119-52404-5
Optional	Robotics, Vision and Control: Fundamental Algorithms In MATLAB	Peter Corke	978-3319544120
Optional	Springer Handbook of Robotics	Bruno Siciliano, Oussama Khatib	978-3-540-30301-5
Optional	A Mathematical Introduction to Robotic Manipulation	Richard M. Murray, Zexiang Li, S. Shankar Sastry	978-0849379819
Optional	Introduction to Autonomous Mobile Robots	Roland Siegwart, Illah Reza Nourbakhsh, Davide Scaramuzza	9780262015356
Optional	Control of Robot Manipulators in Joint Space	Rafael Kelly , Victor Santibáñez Davila , Antonio Loría	978-1-85233-999-9
Optional	Introduction to Modeling and Simulation: A Systems Approach	Mark W. Spong	978-1-119-98288-3
Optional	Linear System Theory and Design	Chi-Tsong Chen	9780199964543
Optional	Applied Nonlinear Control	Jean-Jacques Slotine, Weiping Li	978-0130408907



**APPLIED
NONLINEAR
CONTROL**

Jean-Jacques E. Slotine
Weiping Li

Other Course Materials

1. Webots(<https://cyberbotics.com/>).
2. Visual Studio Code (<https://code.visualstudio.com/>)
3. Robotics Toolbox (<https://github.com/petercorke/robotics-toolbox-python/>) for Python

Grading Criteria

- Assignments – 40%
- Midterm Exam – 30%
- Final Exam – 30%

Grading Criteria

GRADE	PERCENTAGE
A	91-100
B	80-90.9
C	70-79.9
D	60-69.9
F	Below 60

Assignments

- Tasks include readings, research, exercises, and coding challenges aligned with lecture topics.
- Designed to reinforce understanding and provide hands-on experience with theoretical concepts.
- Some assignments may offer **extra credit**, evaluated on complexity and effort, to boost final grades.

Class Guidelines

1. Bring your laptop with Python installed [Visual Studio Code], or use the classroom PCs.
 - Classroom PCs allow running Jupyter Notebook for experiments.
2. Install Webots.
3. It is recommended to use your own laptop with Visual Studio so you can complete assignments at home.
4. Stay updated on **Blackboard** for announcements.
5. This and all other slides will be available on GitHub.

Course Communication and Materials

- Please check Blackboard regularly. All course announcements, grades, and course content will be posted there.
- Code, slides, and other supplementary materials will be available at:
- <https://github.com/gfloresc/RoboticsToolbox-Python-Lessons>
- The GitHub link will also be posted on Blackboard.
- The course syllabus will be updated this week.

Any Questions?

Thank you for your attention!

We will begin the course content in our next class session.